

ATmega128 Final Exam – Integrated Theory & Calculations

1. MCU Architecture & Instruction Model

Harvard Architecture

- Program memory and data memory are physically separate
- Enables simultaneous instruction fetch and data access
- Used by AVR / ATmega128

Exam trigger phrase: “Physically separates program and data memory” → Harvard

Load-Store (RISC) Architecture

- Only LOAD / STORE access memory
- All arithmetic and logic operations occur in registers

Exam contrast: Register-memory architecture allows ALU to operate directly with memory

2. I/O Addressing Methods

Memory-Mapped I/O

- I/O devices share address space with RAM
- Accessed as memory locations

I/O-Mapped I/O

- Separate address space for I/O
- Uses special I/O instructions

Exam rule: “Separate address space” → I/O-mapped I/O; “Accessed as memory” → Memory-mapped I/O

3. Interrupts, Polling, and Function Calls

Interrupt vs Function Call

Interrupt: Hardware / software event, Asynchronous

Function Call: Instruction-triggered, Synchronous

Exam answer: Interrupts are event-triggered

Interrupt vs Polling

Interrupt: Efficient CPU usage, Event-driven, Low latency

Polling: Busy waiting, Periodic, Latency depends on interval

Calculation: Polling interval = 10 μ s → Max latency = 10 μ s

4. UART Communication

UART Frame

- Typical frame: 1 start + 8 data + 1 stop = 10 bits
- Some frames include parity → 11 bits

UART Timing

$$T_{\text{bit}} = 1 / \text{Baud}$$

$$T_{\text{frame}} = \text{Bits} \times T_{\text{bit}}$$

Example: 19200 baud → $T_{\text{bit}} = 52.08 \mu\text{s}$, $T_{\text{frame}} = 0.52 \text{ ms}$

UART Throughput

$$\text{Bytes/sec} = \text{Baud} / \text{Bits per frame}$$

Example: 115200 baud → 11520 bytes/sec

Theory: Interrupt-driven UART provides higher throughput

5. Timers (ATmega128)

Timer Modes: Normal (overflow), CTC (compare reset), PWM, ICP

Not a timer application: PPM

8-bit Timer Overflow

$$T_{\text{ov}} = (256 \times \text{Prescaler}) / f_{\text{CPU}}$$

Example: 16 MHz, prescaler 64 → $T_{\text{ov}} = 1.024 \text{ ms}$

6. PWM (Pulse Width Modulation)

PWM controls average voltage; motor speed $\propto$ duty cycle

$$T = 1 / f$$

$$T_{\text{ON}} = D \times T$$

$$T_{\text{OFF}} = (1 - D) \times T$$

Example: 2 kHz, 40% → $T_{\text{ON}} = 0.2 \text{ ms}$

Example: 1 kHz, 75% → $T_{\text{OFF}} = 0.25 \text{ ms}$

7. ADC (Analog-to-Digital Converter)

Step size: $\Delta V = V_{\text{ref}} / 2^n$

Example: 10-bit, 5 V → 4.88 mV

Input voltage: $V_{\text{in}} = \text{Code} \times \Delta V$

Example: 8-bit, 2.56 V, code 128 → 1.28 V

SAR ADC requires n clock cycles for n-bit resolution

8. DAC (Digital-to-Analog Converter)

$$V_{\text{out}} = (\text{Code} / 2^n) \times V_{\text{ref}}$$

Example: 12-bit, 3.3 V, code 2048 $\rightarrow$ 1.65 V

Linear scaling: 512 $\rightarrow$ 2.5 V, 256 $\rightarrow$ 1.25 V

9. Stepper Motors

Step angle: $\theta = 360^\circ / \text{steps per revolution}$

Example: 400 steps/rev $\rightarrow$ 0.9°

Pulse frequency: $f = (\text{RPM} \times \text{steps/rev}) / 60$

Example: 120 RPM, 200 steps/rev $\rightarrow$ 400 Hz

10. Input Capture (ICP)

$T = \text{Ticks} / f_{\text{timer}}$

Example: 5000 ticks at 2 MHz $\rightarrow$ 2.5 ms

11. Interrupt Vector Addressing

$\text{Address} = \text{Base} + (\text{Vector\#} \times \text{Size})$

Example: Base 0x0000, size 2 bytes, vector #10 $\rightarrow$ 0x0014

12. NRZI Encoding

Rule: 1 $\rightarrow$ toggle, 0 $\rightarrow$ no change

Example: 1010 (initial LOW) $\rightarrow$ HIGH, HIGH, LOW, LOW

13. RTOS Fundamentals

Core modules: Scheduler, Timer, Task management, Communication

NOT a module: Priority (task attribute)

14. Final Exam Strategy

Always check: resolution, frame size, prescaler, ON/OFF time, units

AVR = Harvard + RISC + Load-Store

UART usually assumes 10 bits per frame